

Week 8: Fluids

Fluids Summary

- **Fluids** are states of matter characterized by a lack of long range order. They are characterized by their **density** ρ and their **compressibility**. Liquids such as water are (typically) relatively incompressible; gases can be significantly compressed. Fluids have other characteristics, for example viscosity (how strongly a fluid communicates *shear* stress from a boundary into its bulk volume) and adhesion (how “wet” water is, that is, how strongly it binds to material that is in contact with it at a confining surface).
- **Pressure** is the force per unit area exerted by a fluid on its surroundings:

$$P = F/A \quad (8.1)$$

Its SI units are *pascales* where 1 pascal = 1 newton/meter squared. Pressure is also measured in “atmospheres” (the pressure of air at or near sea level) where 1 atmosphere $\approx 10^5$ pascals. The pressure in an incompressible fluid varies with depth according to:

$$P = P_0 + \rho g D \quad (8.2)$$

where P_0 is the pressure at the top and D is the depth.

- **Pascal’s Principle** Pressure applied to a fluid is transmitted undiminished to all points of the fluid.
- **Archimedes’ Principle** The buoyant force on an object

$$F_b = \rho g V_{\text{disp}} \quad (8.3)$$

where frequency V_{disp} is the volume of fluid displaced by an object.

- **Conservation of Flow** We will study only steady/laminar flow in the absence of turbulence and viscosity.

$$I = A_1 v_1 = A_2 v_2 \quad (8.4)$$

where I is the **flow**, the volume per unit time that passes a given point in e.g. a pipe.

- For a circular smooth round pipe of length L and radius r carrying a fluid in **laminar flow** with **dynamical viscosity**¹²⁶ μ , the flow is related to the pressure difference across the pipe by the **resistance** R :

$$\Delta P = IR \quad (8.5)$$

It is worth noting that this is the fluid-flow version of **Ohm's Law**, which you will learn next semester if you continue. We will generally omit the modifier “dynamical” from the term viscosity in this course, although there is actually another, equivalent measure of viscosity called the **kinematic viscosity**, $\nu = \mu/\rho$. The primary difference is the units – μ has the SI units of pascal-seconds where ν has units of meters square per second.

- The resistance R is given by the follow formula:

$$R = \frac{8L\mu}{\pi r^4} \quad (8.6)$$

and the flow equation above becomes **Poiseuille's Law**¹²⁷:

$$I = \frac{\Delta P}{R} = \frac{\pi r^4 \Delta P}{8L\mu} \quad (8.7)$$

The key facts from this series of definitions are that flow increases linearly with pressure (so to achieve a given e.g. perfusion in a system of capillaries one requires a sufficient pressure difference across them), increases with the **fourth power** of the radius of the pipe (which is why narrowing blood vessels become so dangerous past a certain point) and decreases with the length (longer blood vessels have a greater resistance).

- If we **neglect resistance** (an idealization roughly equivalent to neglecting friction) and consider the flow of fluid in a closed pipe that can e.g. go up and down, the **work-mechanical energy theorem** per unit volume of the fluid can be written as **Bernoulli's Equation**:

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant} \quad (8.8)$$

- **Venturi Effect** At constant height, the pressure in a fluid *decreases* as the velocity of the fluid *increases* (the work done by the pressure difference is what speeds up the fluid!). This is responsible for e.g. the lift of an airplane wing and the force that makes a spinning baseball or golf ball curve.

¹²⁶Wikipedia: <http://www.wikipedia.org/wiki/viscosity>. We will defer any actual statement of how viscosity is related to forces until we cover shear stress in a couple of weeks. It's just too much for now. Oh, and sorry about the symbol. Yes, we already have used μ for e.g. static and kinetic friction. Alas, we will use μ for still more things later. Even with both greek and roman characters to draw on, there just aren't enough characters to cover all of the quantities we want to algebraically work with, so you have to get used to their reuse **in different contexts** that hopefully make them easy enough to keep straight. I decided that it is better to use the accepted symbol in this textbook rather than make one up myself or steal a character from, say, Urdu or a rune from Ancient Norse.

¹²⁷Wikipedia: [http://www.wikipedia.org/wiki/Hagen-Poiseuille equation](http://www.wikipedia.org/wiki/Hagen-Poiseuille%20equation). The derivation of this result isn't *horribly* difficult or hard to understand, but it is long and beyond the scope of this course. Physics and math majors are encouraged to give it a peek though, if only to learn where it comes from.

- **Torricelli's Rule:** If a fluid is flowing through a very small hole (for example at the bottom of a large tank) then the velocity of the fluid at the large end can be neglected in Bernoulli's Equation. In that case the exit speed is the same as the speed of a mass dropped the same distance:

$$v = \sqrt{2gH} \quad (8.9)$$

where H is the depth of the hole relative to the top surface of the fluid in the tank.

8.1: General Fluid Properties

Fluids are the generic name given to two states of matter, liquids and gases¹²⁸ characterized by a lack of long range order and a high degree of mobility at the molecular scale. Let us begin by visualizing fluids microscopically, since we like to build our understanding of matter from the ground up.

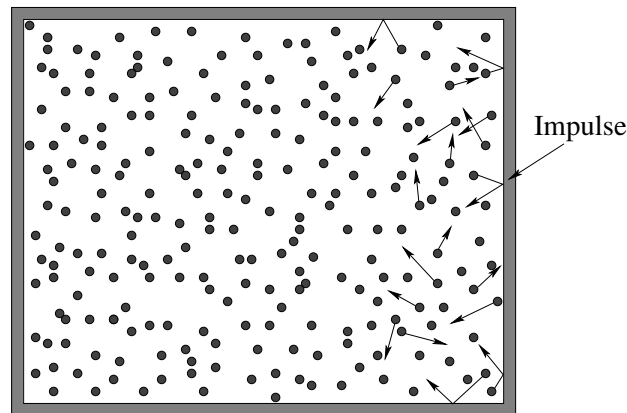


Figure 101: A large number of atoms or molecules are confined within in a “box”, where they bounce around off of each other and the walls. They exert a *force* on the walls equal and opposite the the *force* the walls exert on them as the collisions more or less elastically reverse the particles' momenta perpendicular to the walls.

In figure 101 we see a highly idealized picture of what we might see looking into a tiny box full of gas. Many particles all of mass m are *constantly moving* in random, constantly changing directions (as the particles collide with each other and the walls) with an average kinetic energy related to the temperature of the fluid. Some of the particles (which might be atoms such as helium or neon or molecules such as H_2 or O_2) happen to be close to the walls of the container and moving in the right direction to bounce (elastically) off of those walls.

When they do, their *momentum* perpendicular to those walls is *reversed*. Since *many, many* of these collisions occur each second, there is a nearly continuous momentum transfer between the walls and the gas and the gas and the walls. This transfer, per unit time,

¹²⁸We will not concern ourselves with “plasma” as a possible fourth state of matter in this class, viewing it as just an “ionized gas” although a very dense plasma might well be more like a liquid. Only physics majors and perhaps a few engineers are likely to study plasmas, and you have plenty of time to figure them out after you have learned some electromagnetic theory.

becomes the *average force* exerted by the walls on the gas and the gas on the walls (see the problem in Week 4 with beads bouncing off of a pan).

Eventually, we will transform this simple picture into the **Kinetic Theory of Gases** and use it to derive the venerable **Ideal Gas Law** (physicist style)¹²⁹:

$$PV = Nk_bT \quad (8.10)$$

but for now we will ignore the role of temperature and focus more on understanding the physical characteristics of the fluid such as its **density**, the idea of **pressure itself** and the **force** exerted by fluids on themselves (internally) and on anything the fluid presses upon along the lines of the particles above and the walls of the box.

8.1.1: Pressure

As noted above, the walls of the container exert an average force on the fluid molecules that confine them by reversing their perpendicular momenta in collisions. The total momentum transfer is proportional to the number of molecules that collide per unit time, and this in turn is (all things being equal) clearly proportional to the area of the walls. Twice the surface area – when confining the same number of molecules over each part – has to exert twice the force as twice the number of collisions occur per unit of time, each transferring (on average) the same impulse. It thus makes sense, when considering fluids, to describe the forces that confine and act on the fluids in terms of **pressure**, defined to be the **force per unit area** with which a fluid pushes on a confining wall or the confining wall pushes on the fluid:

$$P = \frac{F}{A} \quad (8.11)$$

Pressure gets its own SI units, which clearly must be Newtons per square meter. We give these units their own name, **Pascals**:

$$1 \text{ Pascal} = \frac{\text{Newton}}{\text{meter}^2} \quad (8.12)$$

A Pascal is a tiny unit of pressure – a Newton isn't very big, recall (one kilogram weighs roughly ten Newtons or 2.2 pounds) so a Pascal is the weight of a quarter pound spread out over a square meter. Writing out "pascal" is a bit cumbersome and you'll see it sometimes abbreviated **Pa** (with the usual power-of-ten modifications, kPa, MPa, GPa, mPa and so on).

A more convenient measure of pressure in our everyday world is a form of the unit called a **bar**:

$$1 \text{ bar} = 10^5 \text{ Pa} = 100 \text{ kPa} \quad (8.13)$$

As it happens, the average air pressure at sea level is *very nearly* 1 bar, and varies by at most a few percent on either side of this. For that reason, air pressure in the modern world is generally reported on the scale of **millibars**, for example you might see air pressure given as 959 mbar (characteristic of the low pressure in a major storm such as a hurricane), 1023 mbar (on a fine, sunny day).

¹²⁹Wikipedia: http://www.wikipedia.org/wiki/Ideal_Gas_Law.

The mbar is probably the “best” of these units for describing everyday air pressure (and its temporal and local and height variation without the need for a decimal or power of ten), with Pascals being an equally good and useful general purpose arbitrary precision unit

The symbol **atm** stands for **one standard atmosphere**. The connection between atmospheres, bars, and pascals is:

$$1 \text{ standard atmosphere} = 101.325 \text{ kPa} = 1013.25 \text{ mbar} \quad (8.14)$$

Note that *real* air pressure at sea level is most unlikely to be this exact value, and although this pressure is often referred to in textbooks and encyclopedias as “the average air pressure at sea level” this is not, in fact, the case. The extra significant digits therefore refer only to a fairly arbitrary value (in pascals) historically related to the original definition of a standard atmosphere in terms of “millimeters of mercury” or *torr*:

$$1 \text{ standard atmosphere} = 760.00 \text{ mmHg} = 760.00 \text{ torr} \quad (8.15)$$

that is of no practical or immediate use. All of this is discussed in some detail in the section on barometers below.

In this class we will use the simple rule $1 \text{ bar} \approx 1 \text{ atm}$ to avoid having to divide out the extra digits, just as we approximated $g \approx 10$ when it is really closer to 9.8. This rule is more than adequate for nearly all purposes and makes pressure arithmetic something you can often do with fingers and toes or the back of an envelope, with around a 1% error if somebody actually gave a pressure in atmospheres with lots of significant digits instead of the superior pascal or bar SI units.

Note well: in the field of medicine **blood pressures** are given in mm of mercury (or torr) by long standing tradition (largely because for at least a century blood pressure was measured with a mercury-based sphygmomanometer). This is discussed further in the section below on the human heart and circulatory system. These can be converted into atmospheres by dividing by 760, remembering that one is measuring the *difference* between these pressures and the standard atmosphere (so the actual blood pressure is always *greater* than one atmosphere).

Pressure isn’t only exerted at the *boundaries* of fluids. Pressure also describes the internal transmission of forces *within* a fluid. For example, we will soon ask ourselves “Why don’t fluid molecules all fall to the ground under the influence of gravity and stay there?” The answer is that (at a sufficient temperature) the *internal pressure* of the fluid suffices to *support* the fluid above upon the back (so to speak) of the fluid below, all the way down to the ground, which of course has to support the weight of the entire column of fluid. Just as “tension” exists in a stretched string at all points along the string from end to end, so the pressure within a fluid is well-defined at all points from one side of a volume of the fluid to the other, although in neither case will the tension or pressure in general be *constant*.

8.1.2: Density

As we have done from almost the beginning, let us note that even a very tiny volume of fluid has many, many atoms or molecules in it, at least under ordinary circumstances in

our everyday lives. True, we can work to create a *vacuum* – a volume that has relatively *few* molecules in it per unit volume, but it is almost impossible to make that number zero – even the hard vacuum of outer space has on average one molecule per cubic meter or thereabouts¹³⁰. We live at the bottom of a gravity well that confines our *atmosphere* – the air that we breathe – so that it forms a relatively thick soup that we move through and breathe with order of *Avogadro's Number* (6×10^{23}) molecules per *liter* – hundreds of billions of billions per cubic centimeter.

At this point we cannot possibly track the motion and interactions of all of the individual molecules, so we *coarse grain* and *average*. The coarse graining means that we once again consider volumes that are large relative to the sizes of the atoms but small relative to our macroscopic length scale of meters – cubic millimeters or cubic microns, for example – that are large enough to contain many, many molecules (and hence a well defined *average* number of molecules) but small enough to treat like a differential volume for the purposes of using calculus to add things up.

We could just count molecules in these tiny volumes, but the properties of *oxygen* molecules and *helium* molecules might well be very different, so the molecular count alone may not be the most useful quantity. Since we are interested in how forces might act on these small volumes, we need to know their mass, and thus we define the *density* of a fluid to be:

$$\rho = \frac{dm}{dV}, \quad (8.16)$$

the **mass per unit volume** we are all familiar with from our discussions of the center of mass of continuous objects and moments of inertia of rigid objects.

Although the definition itself is the same, the density of a fluid behaves in a manner that is similar, but not quite identical in its properties, to the density of a solid. The density of a fluid usually varies *smoothly* from one location to another, because an excess of density in one place will spread out as the molecules travel and collide to smooth out, on average. The particles in some fluids (or almost any fluid at certain temperatures) are “sticky”, or ***strongly interacting***, and hence the fluid coheres together in clumps where the particles are mostly touching, forming a *liquid*. In other fluids (or all fluids at higher temperatures) the molecules move so fast that they do *not* interact much and spend most of their time relatively far apart, forming a *gas*.

A gas spreads itself out to fill any volume it is placed in, subject only to forces that confine it such as the walls of containers or gravity. It assumes the shape of containers, and forms a (usually nearly spherical) layer of atmosphere around planets or stars when confined by gravity. Liquids also spread themselves out to some extent to fill containers they are placed in or volumes they are confined to by a mix of surface forces and gravity, but they also have the property of *surface tension* that can permit a liquid to exert a force of confinement on *itself*. Hence water fills a glass, but water also forms nearly spherical droplets when falling freely as surface tension causes the droplet to minimize its surface area relative to its volume, forming a sphere.

¹³⁰Wikipedia: <http://www.wikipedia.org/wiki/Vacuum>. Vacuum is, of course, “nothing”, and if you take the time to read this Wikipedia article on it you will realize that even *nothing* can be pretty amazing. In man-made vacuums, there are nearly always as many as hundreds of molecules per cubic centimeter.

Surface chemistry or surface adhesion can also exert forces on fluids and initiate things like *capillary flow* of e.g. water up into very fine tubes, drawn there by the surface interaction of the hydrophilic walls of the tube with the water. Similarly, hydrophobic materials can actually repel water and cause water to bead up instead of spreading out to wet the surface. We will largely ignore these phenomena in this course, but they are very interesting and are actually *useful* to physicians as they use pipettes to collect fluid samples that draw themselves up into sample tubes as if by magic. It's not magic, it's just physics.

8.1.3: Compressibility

A major difference between fluids and solids, and liquids and gases within the fluids, is the *compressibility* of these materials. Compressibility describes how a material responds to changes in *pressure*. Intuitively, we expect that if we change the volume of the container (making it smaller, for example, by pushing a piston into a confining cylinder) while holding the amount of material inside the volume constant we will change the pressure; a smaller volume makes for a larger pressure. Although we are not quite prepared to derive and fully justify it, it seems at least *reasonable* that this can be expressed as a simple *linear relationship*:

$$\Delta P = -B \frac{\Delta V}{V} \quad (8.17)$$

Pressure up, volume down and vice versa, where the amount it goes up or down is related, not unreasonably, to the total volume that was present in the first place. The constant of proportionality B is called the **bulk modulus** of the material, and it is very much like (and closely related to) the *spring constant* in Hooke's Law for springs.

Note well that we haven't really specified *yet* whether the "material" is solid, liquid or gas. All three of them have densities, all three of them have bulk moduli. Where they differ is in the *qualitative* properties of their compressibility.

Solids are typically *relatively* incompressible (large B), although there are certainly exceptions. They have long range order – all of the molecules are packed and tightly bonded together in structures and there is usually very little free volume. Atoms themselves violently oppose being "squeezed together" because of the **Pauli exclusion principle** that forbids electrons from having the same set of quantum numbers as well as straight up **Coulomb repulsion** that you will learn about next semester.

Liquids are also relatively incompressible (large B). They differ from solids in that they lack long range order. All of the molecules are constantly moving around and any small "structures" that appear due to local interaction are short-lived. The molecules of a liquid are close enough together that there is often significant physical and chemical interaction, giving rise to surface tension and wetting properties – especially in water, which is (as one sack of water speaking to another) an amazing fluid!

Gases are in contrast quite *compressible* (small B). One can usually squeeze gases smoothly into smaller and smaller volumes, until they reach the point where the molecules are basically all touching and the gas converts to a liquid! Gases per se (especially hot gases) usually remain "weakly interacting" right up to where they become a liquid, although the correct (non-ideal) equation of state for a real gas often displays features that are the

results of moderate interaction, depending on the pressure and temperature.

Water¹³¹ is, as noted, a remarkable liquid. H_2O is a polar molecules with a permanent dipole moment, so water molecules are very strongly interacting, both with each other and with other materials. It organizes itself quickly into a state of relative order that is **very incompressible**. The bulk modulus of water is 2.2×10^9 Pa, which means that even deep in the ocean where pressures can be measured in the tens of millions of Pascals (or hundreds of atmospheres) the density of water only varies by a few percent from that on the surface. Its density varies much more rapidly with *temperature* than with pressure¹³². We will idealize water by considering it to be *perfectly incompressible* in this course, which is close enough to true for nearly any mundane application of hydraulics that you are most unlikely to ever observe an exception that matters.

8.1.4: Viscosity and fluid flow

Fluids, whether liquid or gas, have some internal “stickiness” that resists the relative motion of one part of the fluid compared to another, a kind of internal “friction” that tries to equilibrate an entire body of fluid to move together. They also interact with the walls of any container in which they are confined. The **viscosity** of a fluid (symbol μ) is a measure of this internal friction or stickiness. Thin fluids have a low viscosity and flow easily with minimum resistance; thick sticky fluids have a high viscosity and resist flow.

Fluid, when flowing through (say) a cylindrical pipe tends to organize itself in one of two very different ways – a state of *laminar flow* where the fluid at the very edge of the flowing volume is at rest where it is in contact with the pipe and the speed concentrically and symmetrically increases to a maximum in the center of the pipe, and *turbulent flow* where the fluid tumbles and rolls and forms eddies as it flows through the pipe. Turbulence and flow and viscosity are properties that will be discussed in more detail below.

8.1.5: Properties Summary

To summarize, fluids have the following properties that you should conceptually and intuitively understand and be able to use in working fluid problems:

- They usually assume the shape of any vessel they are placed in (exceptions are associated with confinement due to gravity and surface effects such as surface tension and how well the fluid adheres to the surface in question).
- They are characterized by a mass per unit volume *density* ρ .
- They exert a *pressure* P (force per unit area) on themselves and any surfaces they are in contact with.

¹³¹Wikipedia: http://www.wikipedia.org/wiki/Properties_of_Water. As I said, water is amazing. This article is well worth reading just for fun.

¹³²A fact that impacts my beer-making activities quite significantly, as the specific gravity of hot wort fresh off of the boil is quite different from the specific gravity of the same wort cooled to room temperature. The specific gravity of the wort is related to the sugar content, which is ultimately related to the alcohol content of the fermented beer. Just in case this interests you...

- The pressure can vary according to the dynamic and static properties of the fluid.
- The fluid has a measure of its “stickiness” and resistance to flow called *viscosity*. Viscosity is the internal friction of a fluid, more or less. We will treat fluids as being “ideal” and ignore viscosity in this course.
- Fluids are *compressible* – when the pressure in a fluid is increased, its volume decreases according to the relation:

$$\Delta P = -B \frac{\Delta V}{V} \quad (8.18)$$

where B is called the **bulk modulus** of the fluid (the equivalent of a spring constant). An alternative formulation of this equation uses the **compressibility** β :

$$\beta = -\frac{1}{V} \left(\frac{dV}{dP} \right) = \frac{1}{B}$$

- Fluids where β is a very small number (so large changes in pressure create only tiny changes in fractional volume) are called **incompressible** and their density is roughly constant. Water is an example of an incompressible fluid, as are most liquids. Gases have a higher compressibility and pressure changes can produce a large change in occupied volume (and hence density).
- Below a critical speed, the dynamic flow of a moving fluid tends to be laminar, where every bit of fluid moves parallel to its neighbors in response to pressure differentials and around obstacles. Above that speed it becomes turbulent flow. Turbulent flow is quite difficult to treat mathematically and is hence beyond the scope of this introductory course – we will restrict our attention to ideal fluids either static or in laminar flow.

We will now use these general properties and definitions, plus our existing knowledge of physics, to deduce a number of important properties of and laws pertaining to *static fluids*, fluids that are in static equilibrium.

Static Fluids

8.1.6: Pressure and Confinement of Static Fluids

In figure 102 we see a box of a fluid that is confined within the box by the rigid walls of the box. We will imagine that this particular box is in “free space” far from any gravitational attractor and is therefore at rest with no external forces acting on it. We know from our intuition based on things like cups of coffee that no matter how this fluid is initially stirred up and moving *within* the container, after a very long time the fluid will damp down any initial motion by interacting with the walls of the container and arrive at *static equilibrium*.¹³³

¹³³This state will also entail *thermodynamic equilibrium* with the box (which must be at a uniform temperature) and hence the fluid in this particular non-accelerating box has a uniform density.

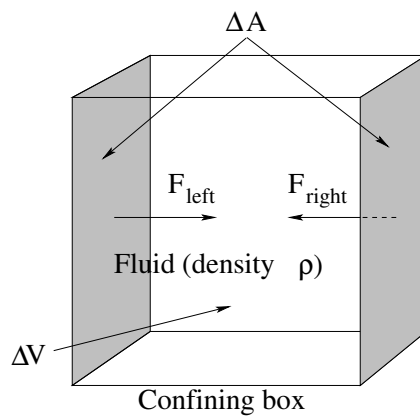


Figure 102: A fluid in static equilibrium confined to a sealed rectilinear box in zero gravity.

A fluid in static equilibrium has the property that every single tiny chunk of volume in the fluid has to *independently* be in *force equilibrium* – the total force acting on the differential volume chunk must be *zero*. In addition the net torques acting on all of these differential subvolumes must be zero, and the fluid must be at rest, neither translating nor rotating. Fluid rotation is more complex than the rotation of a static object because a fluid can be *internally* rotating even if all of the fluid in the outermost layer is in contact with a contain and is *stationary*. It can also be *turbulent* – there can be lots of internal eddies and swirls of motion, including some that can exist at very small length scales and persist for fair amounts of time. We will idealize all of this – when we discuss static properties of fluids we will assume that all of this sort of internal motion has disappeared.

We can now make a few very simple observations about the forces exerted by the walls of the container on the fluid within. First of all the mass of the fluid in the box above is clearly:

$$\Delta M = \rho \Delta V \quad (8.19)$$

where ΔV is the volume of the box. Since it is at rest and remains at rest, the net external force exerted on it (only) by the the box must be zero (see Week 4). We drew a *symmetric* box to make it easy to see that the magnitudes of the forces exerted by opposing walls are equal $F_{\text{left}} = F_{\text{right}}$ (for example). Similarly the forces exerted by the top and bottom surfaces, and the front and back surfaces, must cancel.

The average velocity of the molecules in the box must be zero, but the molecules themselves will generally not be at rest at any nonzero temperature. They will be in a state of constant motion where they *bounce elastically* off of the walls of the box, both giving and receiving an impulse (change in momentum) from the walls as they do. The walls of any box large enough to contain many molecules thus exerts a *nearly continuous* nonzero force that confines any fluid not at zero temperature¹³⁴

From this physical picture we can also deduce an important *scaling property* of the force exerted by the walls. We have deliberately omitted giving any actual dimensions to our box in figure 102. Suppose (as shown) the cross-sectional area of the left and right walls are ΔA originally. Consider now what we expect if we *double the size* of the box

¹³⁴Or at a temperature low enough for the fluid to *freeze* and becomes a *solid*

and at the same time add enough additional fluid for the fluid density to remain the same, making the side walls have the area $2\Delta A$. With twice the area (and twice the volume and twice as much fluid), we have twice as many molecular collisions per unit time on the doubled wall areas (with the same average impulse per collision). The average force exerted by the doubled wall areas therefore *also doubles*.

From this simple argument we can conclude that the average force exerted by any wall is *proportional to the area of the wall*. This force is therefore most naturally expressible in terms of *pressure*, for example:

$$F_{\text{left}} = P_{\text{left}}\Delta A = P_{\text{right}}\Delta A = F_{\text{right}} \quad (8.20)$$

which implies that the *pressure* at the left and right confining walls is the same:

$$P_{\text{left}} = P_{\text{right}} = P \quad (8.21)$$

, and that this pressure describes the force exerted by the fluid on the walls and vice versa. Again, the exact same thing is true for the other four sides.

There is nothing special about our particular choice of left and right. If we had originally drawn a *cubic* box (as indeed we did) we can easily see that the pressure P on *all* the faces of the cube must be the same and indeed (as we shall see more explicitly below) the pressure everywhere in the fluid must be the same!

That's quite a lot of mileage to get out of symmetry and the definition of static equilibrium, but there is one more important piece to get. This last bit involves forces exerted by the wall *parallel* to its surface. On average, there cannot be any! To see why, suppose that one surface, say the left one, exerted a force tangent to the surface itself on the fluid in contact with that surface. An important property of fluids is that *one part of a fluid can move independent of another* so the fluid in at least *some* layer with a finite thickness near the wall would therefore experience a *net* force and would *accelerate*. But this violates our assumption of static equilibrium, so a fluid in *static equilibrium* exerts no tangential force on the walls of a confining container and vice versa.

We therefore conclude that the *direction* of the force exerted by a confining surface with an area ΔA on the fluid that is in contact with it is:

$$\vec{F} = P \Delta A \hat{n} \quad (8.22)$$

where $\hat{n}$ is an inward-directed unit vector *perpendicular to* (normal to) the surface. This final rule permits us to handle the force exerted on fluids confined to *irregular* amoebic blob shaped containers, or balloons, or bags, or – well, us, by our skins and vascular system.

Note well that this says nothing about the tangential force exerted by fluids in *relative motion* to the walls of the confining container. We already know that a fluid moving across a solid surface will exert a *drag force*, and later this week we'll attempt to at least approximately quantify this.

Next, let's consider what happens when we bring this box of fluid¹³⁵ down to Earth and consider what happens to the pressure in a box in *near-Earth gravity*.

¹³⁵It's just a box of rain. I don't know who put it there...

8.1.7: Pressure and Confinement of Static Fluids in Gravity

The principle change brought about by setting our box of fluid down on the ground in a gravitational field (or equivalently, accelerating the box of fluid uniformly in some direction to develop a *pseudo*-gravitational field in the non-inertial frame of the box) is that an additional external force comes into play: The weight of the fluid. A static fluid, confined in some way in a gravitational field, must **support the weight of its many component parts** internally, and of course the box itself must support the weight of the entire mass ΔM of the fluid.

As hopefully you can see if you carefully read the previous section. The only force available to provide the necessary internal support or confinement force is the **variation of pressure within the fluid**. We would like to know how the pressure *varies* as we move up or down in a static fluid so that it supports its own weight.

There are countless reasons that this knowledge is valuable. It is this pressure variation that hurts your ears if you dive deep into the water or collapses submarines if they dive too far. It is this pressure variation that causes your ears to pop as you ride in a car up the side of a mountain or your blood to boil into the vacuum of space if you ride in a rocket all the way out of the atmosphere without a special suit or vehicle that provides a personally pressurized environment. It is this pressure variation that will one day very likely cause you to have varicose veins and edema in your lower extremities from standing on your feet all day – and can help treat/reverse both if you stand in 1.5 meter deep water instead of air. The pressure variation drives water out of the pipes in your home when you open the tap, helps lift a balloon filled with helium, floats a boat but fails to float a rock.

We need to *understand* all of this, whether our eventual goal is to become a physicist, a physician, an engineer, or just a scientifically literate human being. Let's get to it.

Here's the general idea. If we consider a tiny (eventually differentially small) chunk of fluid in force equilibrium, gravity will pull it down and the only thing that can push it up is a *pressure difference* between the top and the bottom of the chunk. By requiring that the force exerted by the pressure difference balance the weight, we will learn how the pressure varies with increasing depth.

For incompressible fluids, this is really all there is to it – it takes only a few lines to derive a lovely formula for the increase in pressure as a function of depth in an incompressible *liquid*.

For gases there is, alas, a small complication. Compressible fluids have densities that *increase* as the pressure increases. This means that boxes of the same size also have *more mass* in them as one descends. More mass means that the pressure difference has to increase, faster, which makes the density/mass greater still, and one discovers (in the end) that the pressure varies *exponentially* with depth. Hence the air pressure drops relatively *quickly* as one goes up from the Earth's surface to very *close* to zero at a height of ten miles, but the atmosphere itself extends for a very long way into space, never quite dropping to "zero" even when one is twenty or a hundred miles high.

As it happens, the calculus for the two kinds of fluids is the *same* up to a given (very important) common point, and then differs, becoming very simple indeed for incompressible fluids and a bit more difficult for compressible ones. Simple solutions suffice to help us

build our conceptual understanding; we will therefore treat incompressible fluids first and **everybody** is responsible for understanding them. Physics majors, math majors, engineers, and people who love a good bit of calculus now and then should probably continue on and learn how to integrate the simple model provided for compressible fluids.

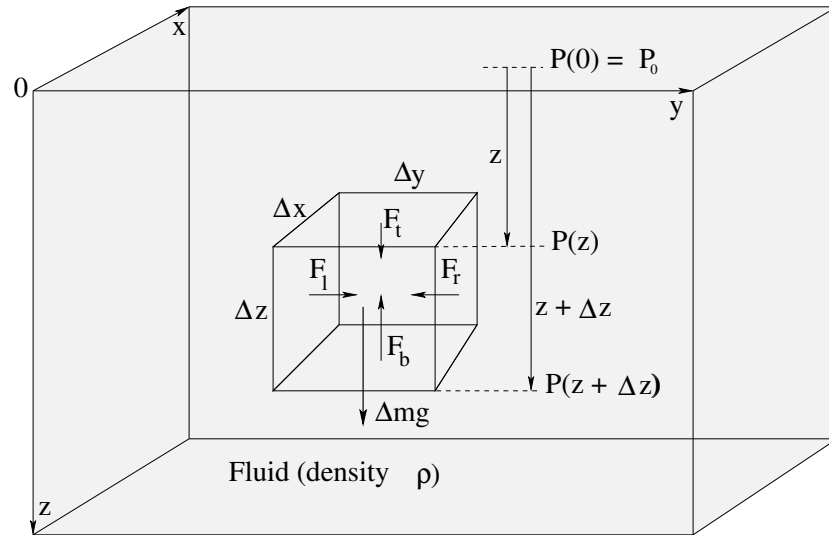


Figure 103: A fluid in static equilibrium confined to a sealed rectilinear box in a near-Earth gravitational field $\vec{g}$. Note well the small chunk of fluid with dimensions $\Delta x, \Delta y, \Delta z$ in the middle of the fluid. Also note that the coordinate system selected has z increasing from the top of the box *down*, so that z can be thought of as the *depth* of the fluid.

In figure 103 a (portion of) a fluid confined to a box is illustrated. The box could be a completely sealed one with rigid walls on all sides, or it could be something like a cup or bucket that is open on the top but where the fluid is still confined there by e.g. atmospheric pressure.

Let us consider a small (eventually infinitesimal) chunk of fluid somewhere in the *middle* of the container. As shown, it has physical dimensions $\Delta x, \Delta y, \Delta z$; its upper surface is a distance z below the origin (where z increases down and hence can represent “depth”) and its lower surface is at depth $z + \Delta z$. The areas of the top and bottom surfaces of this small chunk are e.g. $\Delta A_{tb} = \Delta x \Delta y$, the areas of the sides are $\Delta x \Delta z$ and $\Delta y \Delta z$ respectively, and the volume of this small chunk is $\Delta V = \Delta x \Delta y \Delta z$.

This small chunk is itself in static equilibrium – therefore the forces between any pair of its horizontal sides (in the x or y direction) must cancel. As before (for the box in space) $F_l = F_r$ in magnitude (and opposite in their y -direction) and similarly for the force on the front and back faces in the x -direction, which will always be true if the pressure does not vary *horizontally* with variations in x or y . In the z -direction, however, force equilibrium requires that:

$$F_t + \Delta mg - F_b = 0 \quad (8.23)$$

(where recall, down is positive).

The only possible *source* of F_t and F_b are the **pressure in the fluid itself** which will **vary with the depth** z : $F_t = P(z) \Delta A_{tb}$ and $F_b = P(z + \Delta z) \Delta A_{tb}$. Also, the mass of fluid in the (small) box is $\Delta m = \rho \Delta V$ (using our ritual incantation “the mass of the chunks is...”).

We can thus write:

$$P(z)\Delta x\Delta y + \rho(\Delta x\Delta y\Delta z)g - P(z + \Delta z)\Delta x\Delta y = 0 \quad (8.24)$$

or (dividing by $\Delta x\Delta y\Delta z$ and rearranging):

$$\frac{\Delta P}{\Delta z} = \frac{P(z + \Delta z) - P(z)}{\Delta z} = \rho g \quad (8.25)$$

Finally, we take the limit $\Delta z \rightarrow 0$ and identify the **definition of the derivative** to get:

$$\frac{dP}{dz} = \rho g \quad (8.26)$$

Identical arguments but *without* any horizontal external force followed by $\Delta x \rightarrow 0$ and $\Delta y \rightarrow 0$ lead to:

$$\frac{dP}{dx} = \frac{dP}{dy} = 0 \quad (8.27)$$

as well – P does not vary with x or y as already noted¹³⁶.

In order to find $P(z)$ from this differential expression (which applies, recall, to *any* confined fluid in static equilibrium in a gravitational field) we have to *integrate* it. This integral is very simple if the fluid is incompressible because in that case ρ **is a constant**. The integral isn't *that* difficult if ρ is *not* a constant as implied by the equation we wrote above for the bulk compressibility. We will therefore first do incompressible fluids, then compressible ones.

8.1.8: Variation of Pressure in Incompressible Fluids

In the case of incompressible fluids, ρ is a constant and does not vary with pressure and/or depth. Therefore we can easily multiple $dP/dz = \rho g$ above by dz on both sides and integrate to find P :

$$\begin{aligned} dP &= \rho g dz \\ \int dP &= \int \rho g dz \\ P(z) &= \rho g z + P_0 \end{aligned} \quad (8.28)$$

where P_0 is the *constant of integration* for both integrals, and practically speaking is the pressure in the fluid at zero depth (wherever that might be in the coordinate system chosen).

Example 8.1.1: Barometers

Mercury barometers were originally invented by Evangelista Torricelli¹³⁷ a natural philosopher who acted as Galileo's secretary for the last three months of Galileo's life under house

¹³⁶Physics and math majors and other students of multivariate calculus will recognize that I should probably be using partial derivatives here and establishing that $\vec{\nabla}P = \rho\vec{g}$, where in free space we should instead have had $\vec{\nabla}P = 0 \rightarrow P$ constant.

¹³⁷Wikipedia: http://www.wikipedia.org/wiki/Evangelista_Torricelli.

arrest. The invention was inspired by Torricelli's attempt to solve an important engineering problem. The pump makers of the Grand Duke of Tuscany had built powerful pumps intended to raise water twelve or more meters, but discovered that no matter how powerful the pump, water stubbornly refused to rise more than ten meters into a pipe evacuated at the top.

Torricelli demonstrated that a shorter glass tube filled with mercury, when inverted into a dish of mercury, would fall back into a column with a height of roughly 0.76 meters with a vacuum on top, and soon thereafter discovered that the height of the column fluctuated with the pressure of the outside air pressing down on the mercury in the dish, correctly concluding that water would behave exactly the same way¹³⁸. Torricelli made a number of other important 17th century discoveries, correctly describing the causes of wind and discovering "Torricelli's Law" (an aspect of the Bernoulli Equation we will note below).

In honor of Torricelli, a unit of pressure was named after him. The **torr** is the pressure required to push the mercury in Torricelli's barometer up one millimeter. Because mercury barometers were at one time nearly ubiquitous as the most precise way to measure the pressure of the air, a specific height of the mercury column was the original definition of the standard atmosphere. For better or worse, Torricelli's original observation defined one standard atmosphere to be *exactly* "760 millimeters of mercury" (which is a lot to write or say) or as we would now say, "760 torr"¹³⁹.

Mercury barometers are now more or less banned, certainly from the workplace, because mercury is a potentially toxic heavy metal. In actual fact, *liquid* mercury is not biologically active and hence is not particularly toxic. Mercury vapor *is* toxic, but the amount of mercury vapor emitted by the exposed surface of a mercury barometer at room temperature is well below the levels considered to be a risk to human health by OSHA unless the barometer is kept in a small, hot, poorly ventilated room with someone who works there over years. This isn't all that common a situation, but with all toxic metals we are probably better safe than sorry¹⁴⁰.

At this point mercury barometers are rapidly disappearing everywhere but from the hands of collectors. Their manufacture is banned in the U.S., Canada, Europe, and many other nations. We had a lovely one (probably more than one, but I recall one) in the Duke Physics Department up until sometime in the 90's¹⁴¹, but it was – sanely enough – removed and retired during a renovation that also cleaned up most if not all of the asbestos in the building. Ah, my toxic youth...

¹³⁸You, too, get to solve Torricelli's problem as one of your homework problems, but armed with a lot better understanding.

¹³⁹Not to be outdone, one standard atmosphere (or atmospheric pressures in weather reporting) in the U.S. is often given as 29.92 barleycorn-derived *inches* of mercury instead of millimeters. Sigh.

¹⁴⁰The single biggest risk associated with uncontained liquid mercury (in a barometer or otherwise) is that you can easily spill it, and once spilled it is fairly likely to sooner or later make its way into either *mercury vapor* or **methly mercury**, both of which are biologically active and highly toxic. Liquid mercury itself you could drink a glass of and it would pretty much pass straight through you with minimal absorption and little to no damage *if* you – um – "collected" it carefully and disposed of it properly on the other side.

¹⁴¹I used to work in the small, cramped space with poor circulation where it was located from time to time but never very long at a time and besides, the room was *cold*. But if I seem "mad as a hatter" – mercury nitrate was used in the making of hats and the vapor used to poison the hatters vapor used to poison hat makers – it probably isn't from this...

Still, at one time they were *extremely* common – most ships had one, many households had one, businesses and government agencies had them – knowing the pressure of the air is an important factor in weather prediction. Let's see how they work(ed).

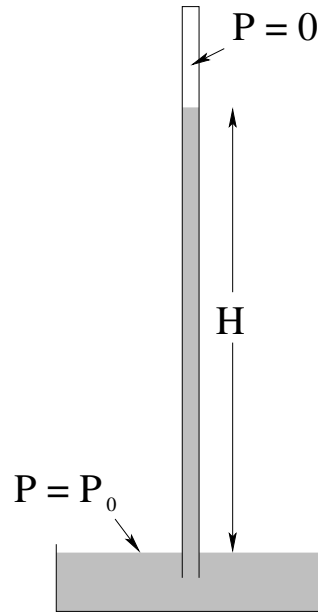


Figure 104: A simple fluid barometer consists of a tube with a vacuum at the top filled with fluid supported by the air pressure outside.

A simple mercury barometer is shown in figure [104](#). It consists of a tube that is completely filled with mercury. Mercury has a specific gravity of 13.534 at a typical room temperature, hence a density of 13534 kg/m^3). The filled tube is then inverted into a small reservoir of mercury (although other designs of the bottom are possible, some with smaller exposed surface area of the mercury). The mercury falls (pulled down by gravity) out of the tube, leaving behind a ***vacuum*** at the top.

We can easily compute the expected height of the mercury column if P_0 is the pressure on the exposed surface of the mercury in the reservoir. In that case

$$P = P_0 + \rho g z \quad (8.29)$$

as usual for an incompressible fluid. Applying this formula to both the top and the bottom,

$$P(0) = P_0 \quad (8.30)$$

and

$$P(H) = P_0 - \rho g H = 0 \quad (8.31)$$

(recall that the upper surface is *above* the lower one, $z = -H$). From this last equation:

$$P_0 = \rho g H \quad (8.32)$$

and one can easily convert the measured height H of mercury above the top surface of mercury in the reservoir into P_0 , the air pressure on the top of the reservoir.

At one standard atmosphere, we can easily determine what a mercury barometer at room temperature will read (the height H of its column of mercury above the level of mercury in the reservoir):

$$P_0 = 13534 \frac{\text{kg}}{\text{m}^3} \times 9.80665 \frac{\text{m}}{\text{sec}^2} \times H = 101325 \text{Pa} \quad (8.33)$$

Note well, we have used the precise SI value of g in this expression, and the density of mercury at “room temperature” around 20°C or 293°K. Dividing, we find the expected height of mercury in a barometer at room temperature and one standard atmosphere is $H = 0.763$ meters or 763 torr

Note that this is not exactly the 760 torr we expect to read for a standard atmosphere. This is because for high precision work one cannot just use *any old* temperature (because mercury has a significant thermal expansion coefficient and was then and continues to be used today in mercury thermometers as a consequence). The unit definition is based on using the density of mercury at 0°C or 273.16°K, which has a specific gravity (according to NIST, the National Institute of Standards in the US) of 13.595. Then the precise connection between SI units and torr follows from:

$$P_0 = 13595 \frac{\text{kg}}{\text{m}^3} \times 9.80665 \frac{\text{m}}{\text{sec}^2} \times H = 101325 \text{Pa} \quad (8.34)$$

Dividing we find the value of H expected at one standard atmosphere:

$$H_{\text{atm}} = 0.76000 = 760.00 \text{ millimeters} \quad (8.35)$$

Note well the precision, indicative of the fact that the SI units for a standard atmosphere *follow* from their definition in torr, not the other way around.

Curiously, this value is invariably given in both textbooks and even the wikipedia article on atmospheric pressure as the *average* atmospheric pressure at sea level, which it almost certainly is not – a spatiotemporal averaging of sea level pressure would have been utterly impossible during Torricelli’s time (and would be difficult today!) and if it was done, could not possibly have worked out to be *exactly* 760.00 millimeters of mercury at 273.16°K.

Example 8.1.2: Variation of Oceanic Pressure with Depth

The pressure on the surface of the ocean is, approximately, by definition, one atmosphere. Water is a highly incompressible fluid with $\rho_w = 1000$ kilograms per cubic meter¹⁴². $g \approx 10$ meters/second². Thus:

$$P(z) = P_0 + \rho_w g z = (10^5 + 10^4 z) \text{ Pa} \quad (8.36)$$

or

$$P(z) = (1.0 + 0.1z) \text{ bar} = (1000 + 100z) \text{ mbar} \quad (8.37)$$

Every ten meters of depth (either way) increases water pressure by (approximately) one atmosphere!

Wow, *that* was easy. This is a *very important rule of thumb* and is actually fairly easy to remember! How about compressible fluids?

¹⁴²Good number to remember. In fact, *great* number to remember.

8.1.9: Variation of Pressure in Compressible Fluids

Compressible fluids, as noted, have a density which *varies* with pressure. Recall our equation for the compressibility:

$$\Delta P = -B \frac{\Delta V}{V} \quad (8.38)$$

If one increases the pressure, one therefore *decreases* occupied volume of any given chunk of mass, and hence increases the density. However, to predict precisely how the density will depend on pressure requires more than just this – it requires a *model* relating pressure, volume and mass.

Just such a model for a compressible gas is provided (for example) by the **Ideal Gas Law**¹⁴³:

$$PV = Nk_bT = nRT \quad (8.39)$$

where N is the number of molecules in the volume V , k_b is Boltzmann's constant¹⁴⁴ n is the number of moles of gas in the volume V , R is the ideal gas constant¹⁴⁵ and T is the temperature in degrees Kelvin (or Absolute)¹⁴⁶. If we assume constant temperature, and convert N to the mass of the gas by multiplying by the molar mass and dividing by Avogadro's Number¹⁴⁷ 6×10^{23} .

(**Aside:** If you've never taken chemistry a lot of this is going to sound like Martian to you. Sorry about that. As always, consider visiting the e.g. Wikipedia pages linked above to learn enough about these topics to get by for the moment, or just keep reading as the *details* of all of this won't turn out to be very important...)

When we do this, we get the following formula for the density of an ideal gas:

$$\rho = \frac{M}{RT}P \quad (8.40)$$

where M is the molar mass¹⁴⁸, the number of kilograms of the gas per mole. Note well that this result is *idealized* – that's why they call it the *Ideal* Gas Law! – and that no real gases are “ideal” for all pressures and temperatures because sooner or later they all become *liquids* or *solids* due to molecular interactions. However, the gases that make up “air” are all reasonably ideal at temperatures in the ballpark of room temperature, and in any event it is worth seeing how the pressure of an ideal gas varies with z to get an *idea* of how air pressure will vary with height. Nature will probably be somewhat different than this prediction, but we ought to be able to make a *qualitatively* accurate model that is also moderately *quantitatively* predictive as well.

As mentioned above, the formula for the derivative of pressure with z is unchanged for compressible or incompressible fluids. If we take $dP/dz = \rho g$ and multiply both sides by

¹⁴³Wikipedia: http://www.wikipedia.org/wiki/Ideal_Gas_Law.

¹⁴⁴Wikipedia: http://www.wikipedia.org/wiki/Boltzmann's_Constant.

¹⁴⁵Wikipedia: http://www.wikipedia.org/wiki/Gas_Constant.

¹⁴⁶Wikipedia: <http://www.wikipedia.org/wiki/Temperature>.

¹⁴⁷Wikipedia: http://www.wikipedia.org/wiki/Avogadro's_Number.

¹⁴⁸Wikipedia: http://www.wikipedia.org/wiki/Molar_Mass.

dz as before and integrate, now we get (assuming a fixed temperature T):

$$\begin{aligned} dP &= \rho g dz = \frac{Mg}{RT} P dz \\ \frac{dP}{P} &= \frac{Mg}{RT} dz \\ \int \frac{dP}{P} &= \frac{Mg}{RT} \int dz \\ \ln(P) &= \frac{Mg}{RT} z + C \end{aligned}$$

We now do the usual¹⁴⁹ – exponentiate both sides, turn the exponential of the sum into the product of exponentials, turn the exponential of a constant of integration into a constant of integration, and match the units:

$$P(z) = P_0 e^{\left(\frac{Mg}{RT}\right)z} \quad (8.41)$$

where P_0 is the pressure at zero *depth*, because (recall!) z is measured positive *down* in our expression for dP/dz .

Example 8.1.3: Variation of Atmospheric Pressure with Height

Using z to describe depth is moderately inconvenient, so let us define the **height h above sea level** to be $-z$. In that case P_0 is (how about that!) *1 Atmosphere*. The molar mass of dry air is $M = 0.029$ kilograms per mole. $R = 8.31$ Joules/(mole-K°). Hence a bit of multiplication at $T = 300^\circ$:

$$\frac{Mg}{RT} = \frac{0.029 \times 10}{8.31 \times 300} = 1.12 \times 10^{-4} \text{ meters}^{-1} \quad (8.42)$$

or:

$$P(h) = 10^5 \exp(-0.00012 h) \text{ Pa} = 1000 \exp(-0.00012 h) \text{ mbar} \quad (8.43)$$

Note well that the temperature of air is *not* constant as one ascends – it drops by a fairly significant amount, even on the absolute scale (and higher still, it *rises* by an even *greater* amount before dropping again as one moves through the layers of the atmosphere. Since the pressure is found from an integral, this in turn means that the exponential behavior itself is rather inexact, but still it isn't a *terrible* predictor of the variation of pressure with height. This equation predicts that air pressure should drop to $1/e$ of its sea-level value of 1000 mbar at a height of around 8000 meters, the height of the so-called **death zone**¹⁵⁰. We can compare the actual (average) pressure at 8000 meters, 356 mbar, to $1000 \times e^{-1} = 368$ mbar. We get remarkably good agreement!

¹⁴⁹Which should be familiar to you both from solving the linear drag problem in Week 2 and from the online Math Review.

¹⁵⁰Wikipedia: http://www.wikipedia.org/wiki/Effects_of_high_altitude_on_humans. This is the height where air pressure drops to where humans are at extreme risk of dying if they climb without supplemental oxygen support – beyond this height hypoxia reduces one's ability to make important and life-critical decisions during the very last, most stressful, part of the climb. Mount Everest (for example) can only be climbed with oxygen masks and some of the greatest disasters that have occurred climbing it and other peaks are associated with a lack of or failure of supplemental oxygen.

This agreement rapidly breaks down, however, and meteorologists actually use a patch-work of formulae (both algebraic and exponential) to give better agreement to the actual variation of air pressure with height as one moves up and down through the various named layers of the atmosphere with the pressure, temperature and even molecular composition of “air” varying all the way. This simple model explains a *lot* of the variation, but its assumptions are not really correct.

8.2: Pascal's Principle and Hydraulics

We note that (from the above) the *general form* of P of a fluid confined to a sealed container has the most general form:

$$P(z) = P_0 + \int_0^z \rho g dz \quad (8.44)$$

where P_0 is the constant of integration or value of the pressure at the reference **depth** $z = 0$. This has an important consequence that forms the basis of *hydraulics*.

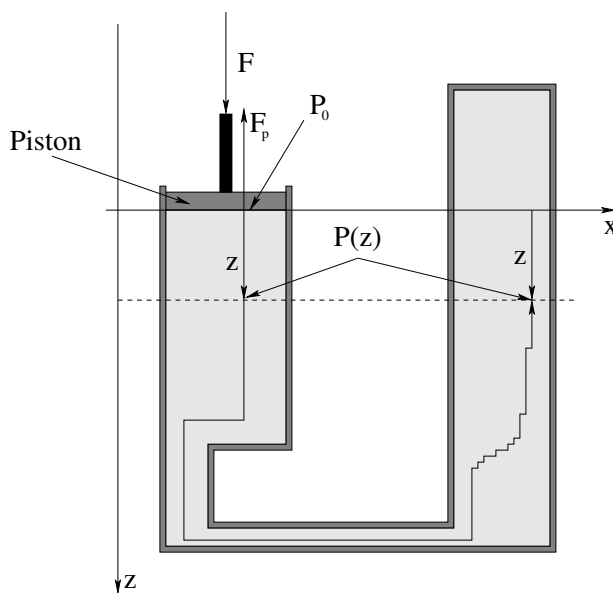


Figure 105: A single piston seated tightly in a frictionless cylinder of cross-sectional area A is used to compress water in a sealed container. Water is incompressible and does not significantly change its volume at $P = 1$ bar (and a constant room temperature) for pressure changes on the order of 0.1-100 bar.

Suppose, then, that we have an **incompressible fluid** e.g. water confined within a sealed container by e.g. a *piston* that can be pushed or pulled on to *increase or decrease* the *confinement pressure* on the surface of the piston. Such an arrangement is portrayed in figure [105](#)

We can push down (or pull back) on the piston with any total downward force F that we like that leaves the system in equilibrium. Since the piston itself is in static equilibrium, the force we push with must be opposed by the pressure in the fluid, which exerts an equal

and opposite upwards force:

$$F = F_p = P_0 A \quad (8.45)$$

where A is the cross sectional area of the piston and where we've put the cylinder face at $z = 0$, which we are obviously free to do. For all practical purposes this means that we can make P_0 "anything we like" within the range of pressures that are unlikely to make water at room temperature change its state or volume do other bad things, say $P = (0.1, 100)$ bar.

The pressure at a depth z in the container is then (from our previous work):

$$P(z) = P_0 + \rho g z \quad (8.46)$$

where $\rho = \rho_w$ if the cylinder is indeed filled with water, but the cylinder could equally well be filled with hydraulic fluid (basically oil, which assists in lubricating the piston and ensuring that it remains "frictionless" while assisting the seal), alcohol, mercury, or any other incompressible liquid.

We recall that the pressure changes *only* when we change our depth. Moving laterally does not change the pressure, because e.g. $dP/dx = dP/dy = 0$. We can always find a path consisting of vertical and lateral displacements from $z = 0$ to any other point in the container – two such points at the same depth z are shown in [105](#) along with a (deliberately ziggy-zaggy [151](#)) vertical/horizontal path connecting them. Clearly these two points **must have the same pressure $P(z)$!**

Now consider the following. Suppose we start with pressure P_0 (so that the pressure at these two points is $P(z)$), but then change F to make the pressure P'_0 and the pressure at the two points $P'(z)$. Then:

$$\begin{aligned} P(z) &= P_0 + \rho g z \\ P'(z) &= P'_0 + \rho g z \\ \Delta P(z) &= P'(z) - P(z) = P'_0 - P_0 = \Delta P_0 \end{aligned} \quad (8.47)$$

That is, the pressure change at depth z does not depend on z *at any point in the fluid!* It depends *only on the change in the pressure exerted by the piston!*

This result is known as **Pascal's Principle** and it holds (more or less) for *any* compressible fluid, not just incompressible ones, but in the case of compressible fluids the piston will move up or down or in or out and the density of the fluid will change and hence the treatment of the integral will be too complicated to cope with. Pascal's Principle is more commonly given in *English words* as:

Any change in the pressure exerted at a given point on a confined fluid is transmitted, undiminished, throughout the fluid.

Pascal's principle is the basis of **hydraulics**. Hydraulics are a kind of fluid-based simple machine that can be used to greatly amplify an applied force. To understand it, consider the following figure:

¹⁵¹Because we can make the zigs and zags *differentially small*, at which point this piecewise horizontal-vertical line becomes an *arbitrary curve* that remain in the fluid. Multivariate calculus can be used to formulate all of these results more prettily, but the *reasoning* behind them is completely contained in the picture and this text explanation.

Example 8.2.1: A Hydraulic Lift

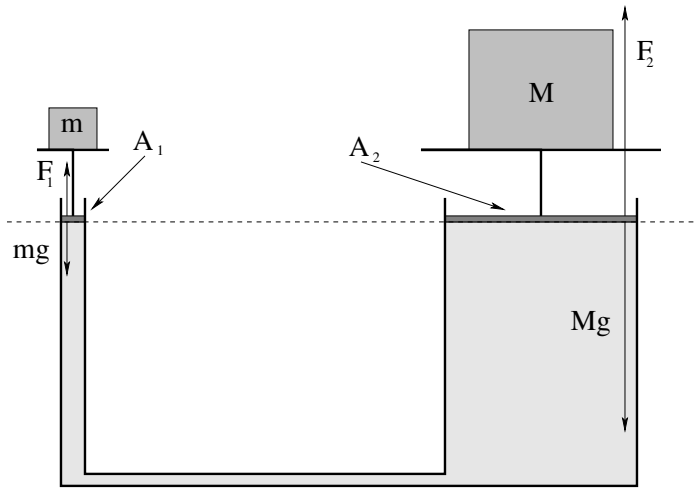


Figure 106: A simple schematic for a hydraulic lift of the sort used in auto shops to lift your car.

Figure 106 illustrates the way we can multiply forces using Pascal's Principle. Two pistons seal off a pair of cylinders connected by a closed tube that contains an incompressible fluid. The two pistons are deliberately given the same height (which might as well be $z = 0$, then, in the figure, although we could easily deal with the variation of pressure associated with them being at different heights since we know $P(z) = P_0 + \rho gz$). The two pistons have cross sectional areas A_1 and A_2 respectively, and support a small mass m on the left and large mass M on the right in static equilibrium.

For them to be in equilibrium, clearly:

$$F_1 - mg = 0 \quad (8.48)$$

$$F_2 - Mg = 0 \quad (8.49)$$

We also/therefore have:

$$F_1 = P_0 A_1 = mg \quad (8.50)$$

$$F_2 = P_0 A_2 = Mg \quad (8.51)$$

Thus

$$\frac{F_1}{A_1} = P_0 = \frac{F_2}{A_2} \quad (8.52)$$

or (substituting and cancelling g):

$$M = \frac{A_2}{A_1} m \quad (8.53)$$

A small mass on a small-area piston can easily balance a *much larger mass on an equally larger area piston!*

Just like a lever, we can balance or lift a large weight with a small one. Also just as was the case with a lever, *there ain't no such thing as a free lunch!* If we try to *lift* (say) a car with a hydraulic lift, we have to move the same volume $\Delta V = A\Delta z$ from under the small

piston (as it descends) to under the large one (as it ascends). If the small one goes down a distance z_1 and the large one goes up a distance z_2 , then:

$$\frac{z_1}{z_2} = \frac{A_2}{A_1} \quad (8.54)$$

The *work* done by the two cylinders thus *precisely balances*:

$$W_2 = F_2 z_2 = F_1 \frac{A_2}{A_1} z_2 = F_1 \frac{A_2}{A_1} z_1 \frac{A_1}{A_2} = F_1 z_1 = W_1 \quad (8.55)$$

The hydraulic arrangement thus transforms pushing a small force through a large distance into a large force moved through a small distance so that the work done **on** piston 1 matches the work done **by** piston 2. No energy is created or destroyed (although in the real world a bit will be lost to heat as things move around) and all is well, quite literally, with the Universe.

This example is pretty simple, but it should suffice to guide you through doing a work-energy conservation problem where (for example) the mass m goes down a distance d (losing gravitational energy) and the mass M goes up a distance D (gaining gravitational energy *while the fluid itself also is net moved up above its former level!* Don't forget that last, tricky bit if you ever have a problem like that!

8.3: Fluid Displacement and Buoyancy

First, a story. Archimedes¹⁵² was, quite possibly, the smartest person who has ever lived (so far). His day job was being the “court magician” in the island kingdom of Syracuse in the third century BCE, some 2200 years ago; in his free time he did things like invent primitive integration, accurately compute pi, invent amazing machines of war and peace, determine the key principles of both statics and fluid statics (including the one we are about to study and the principles of the lever – “Give me put a place to stand and I can move the world!” is a famous Archimedes quote, implying that a sufficiently long lever would allow the small forces humans can exert to move even something as large as the Earth, although yeah, there are a few problems with that that go beyond just a place to stand¹⁵³).

The king (Hieron II) of Syracuse had a problem. He had given a goldsmith a mass of pure gold to make him a **votive crown**, but when the crown came back he had the niggling suspicion that the goldsmith had substituted cheap silver for some of the gold and kept the gold. It was keeping him awake at nights, because if somebody can steal from the king and get away with it (and word gets out) it can only encourage a loss of respect and rebellion.

So he called in his court magician (Archimedes) and gave him the task of determining whether or not the crown had been made by adulterated gold – or else. And oh, yeah –

¹⁵²Wikipedia: <http://www.wikipedia.org/wiki/Archimedes>. A very, very interesting person. I strongly recommend that my students read this short article on this person who came *within a hair* of inventing physics and calculus and starting the Enlightenment some 1900 years before Newton. Scary supergenius polymath guy. Would have won multiple Nobel prizes, a Macarthur “Genius” grant, and so on if alive today. Arguably the smartest person who has ever lived – so far.

¹⁵³The “sound bite” is hardly a modern invention, after all. Humans have always loved a good, pithy statement of insight, even if it isn't actually even approximately true...

you can't melt down the crown and cast it back into a regular shape whose dimensions can be directly compared to the same shape of gold, permitting a direct comparison of their *densities* (the density of pure gold is not equal to the density of gold with an admixture of silver). And don't forget the "or else".

Archimedes puzzled over this for some days, and decided to take a bath and cool off his overheated brain. In those days, baths were large public affairs – you *went* to the baths as opposed to having one in your home – where a filled tub was provided, sometimes with attendants happy to help you wash. As the possibly apocryphal story has it, Archimedes lowered himself into the overfull tub and as he did so, water sloshed out as he *displaced its volume* with his own volume. In an intuitive, instantaneous flash of insight – a "light bulb moment" – he realized that *displacement of a liquid by an irregular shaped solid* can be used to measure its volume, and that such a measurement of displaced volume would allow the king's problem to be solved.

Archimedes then leaped out of the tub and ran naked through the streets of Syracuse (which we can only imagine provided its inhabitants with as much amusement then as it would provide now) yelling "Eureka!", which in Greek means "I have found it!" The test (two possible versions of which are supplied below, one more probable than the other but less instructive for our own purposes) was performed, and revealed that the goldsmith was indeed dishonest and had stolen some of the king's gold. Bad move, goldsmith! We will draw a tasteful veil over the probable painful and messy fate of the goldsmith.

Archimedes transformed his serendipitous discovery of static fluid displacement into an elaborate physical principle that explained *buoyancy*, the tendency of fluids to support all or part of the weight of objects immersed in them.

The fate of Archimedes himself is worth a moment more of our time. In roughly 212 BCE, the Romans invaded Syracuse in the Second Punic War after a two year siege. As legend has it, as the city fell and armed soldiers raced through the streets "subduing" the population as only soldiers can, Archimedes was in his court chambers working on a problem in the geometry of circles, which he had drawn out in the sand boxes that then served as a "chalkboard". A Roman soldier demanded that he leave his work and come meet with the conquering general, Marcus Claudius Marcellus. Archimedes declined, replying with his last words "Do not disturb my circles" and the soldier killed him. Bad move, soldier – Archimedes himself was a major part of the loot of the city and Marcellus had ordered that he was not to be harmed. The fate of the soldier that killed him is unknown, but it wasn't really a very good idea to anger a conquering general by destroying an object or person of enormous value, and I doubt that it was very good.

Anyway, let's see the modern version of Archimedes' discovery and see as well how Archimedes very probably used it to test the crown.

8.3.1: Archimedes' Principle

If you are astute, you will note that figure [107](#) is **exactly like figure [103](#)** above, except that the internal chunk of *fluid* has been replaced by *some other material*. The point is that this replacement does not matter – **the net force exerted on the cube by the fluid is the**

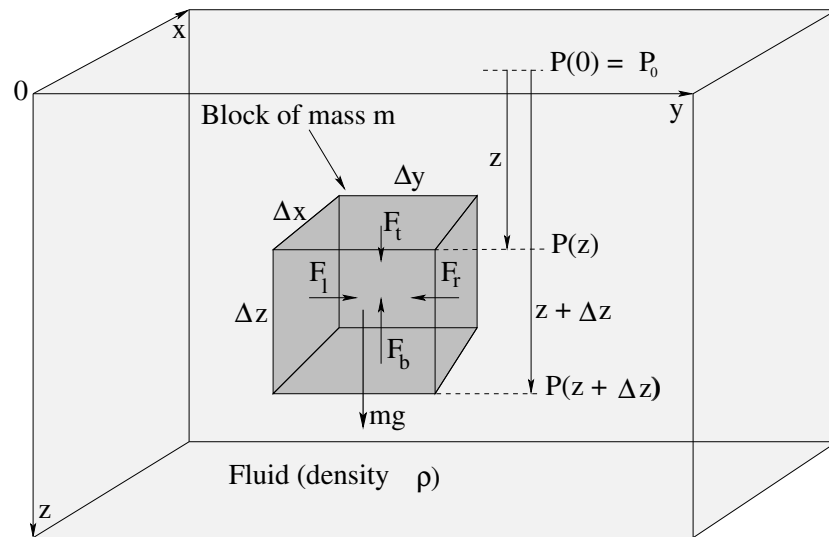


Figure 107: A solid chunk of “stuff” of mass m and the dimensions shown is immersed in a fluid of density ρ at a depth z . The vertical pressure difference in the fluid (that arises as the fluid itself becomes static static) exerts a vertical force on the cube.

same!

Hopefully, that is obvious. The net upward force exerted by the fluid is called the **buoyant force** F_b and is equal to:

$$\begin{aligned}
 F_b &= P(z + \Delta z)\Delta x\Delta y - P(z)\Delta x\Delta y \\
 &= ((P_0 + \rho g(z + \Delta z)) - (P_0 + \rho gz))\Delta x\Delta y \\
 &= \rho g\Delta z\Delta x\Delta y \\
 &= \rho g\Delta V
 \end{aligned} \tag{8.56}$$

where ΔV is the volume of the small block.

The buoyant force is thus the **weight of the fluid displaced** by this single tiny block. This is all we need to show that the same thing is true for an *arbitrary* immersed shape of object.

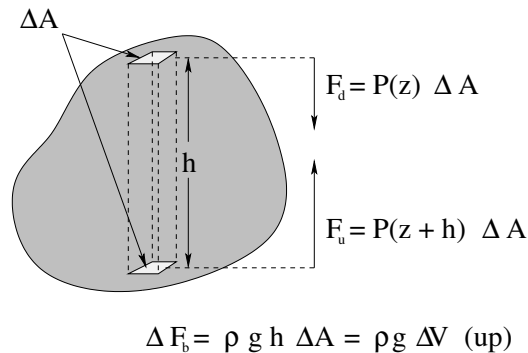


Figure 108: An arbitrary chunk of stuff is immersed in a fluid and we consider a vertical cross section with horizontal ends of area ΔA and height h through the chunk.

In figure [108](#), an arbitrary blob-shape is immersed in a fluid (not shown) of density ρ .

Imagine that we've taken a french-fry cutter and cuts the whole blob into nice rectangular segments, one of which (of length h and cross-sectional area ΔA) is shown. We can trim or average the end caps so that they are all perfectly horizontal by making all of the rectangles arbitrarily small (in fact, differentially small in a moment). In that case the *vertical* force exerted by the fluid on just the two lightly shaded surfaces shown would be:

$$F_d = P(z)\Delta A \quad (8.57)$$

$$F_u = P(z+h)\Delta A \quad (8.58)$$

where we assume the upper surface is at depth z (this won't matter, as we'll see in a moment). Since $P(z+h) = P(z) + \rho gh$, we can find the *net upward buoyant force* exerted on this little cross-section by subtracting the first from the second:

$$\Delta F_b = F_u - F_d = \rho g h \Delta A = \rho g \Delta V \quad (8.59)$$

where the *volume* of this piece of the entire blob is $\Delta V = h \Delta A$.

We can now let $\Delta A \rightarrow dA$, so that $\Delta V \rightarrow dV$, and write

$$F_b = \int dF_b = \int_{V \text{ of blob}} \rho g dV = \rho g V = m_f g \quad (8.60)$$

where $m_f = \rho V$ is the mass of the fluid displaced, so that $m_f g$ is its weight.

That is:

The total buoyant force on the immersed object is the weight of the fluid displaced by the object.

This is really an adequate *proof* of this statement, although if we were really going to be picky we'd use the fact that P doesn't vary in x or y to show that the net force in the x or y direction is zero independent of the shape of the blob, using our differential french-fry cutter mentally in the x direction and then noting that the blob is *arbitrary* in shape and we could have just as easily labelled or oriented the blob with this direction called y so it must be true in *any* direction perpendicular to $\vec{g}$.

This statement – in the English or algebraic statement as you prefer – is known as ***Archimedes' Principle***, although Archimedes could hardly have formulated it quite the way we did algebraically above as he died before he could *quite* finish inventing the calculus and physics.

This principle is enormously important and ubiquitous. Buoyancy is why boats float, but rocks don't. It is why childrens' helium-filled balloons do odd things in accelerating cars. It exerts a subtle force on everything submerged in the air, in water, in beer, in liquid mercury, as long as the fluid itself is either in a gravitational field (and hence has a pressure gradient) or is in an accelerating container with its own "pseudogravity" (and hence has a pressure gradient).

Let's see how Archimedes could have used this principle to test the crown two ways. The first way is very simple and *conceptually* instructive; the second way is more practical to us as it illustrates the way we generally do algebra associated with buoyancy problems.

Example 8.3.1: Testing the Crown I

The tools Archimedes probably had available to him were balance-type scales, as these tools for comparatively measuring the weight were well-known in antiquity. He certainly had vessels he could fill with water. He had thread or string, he had the crown itself, and he had access to pure gold from the king's treasury (at least for the duration of the test).

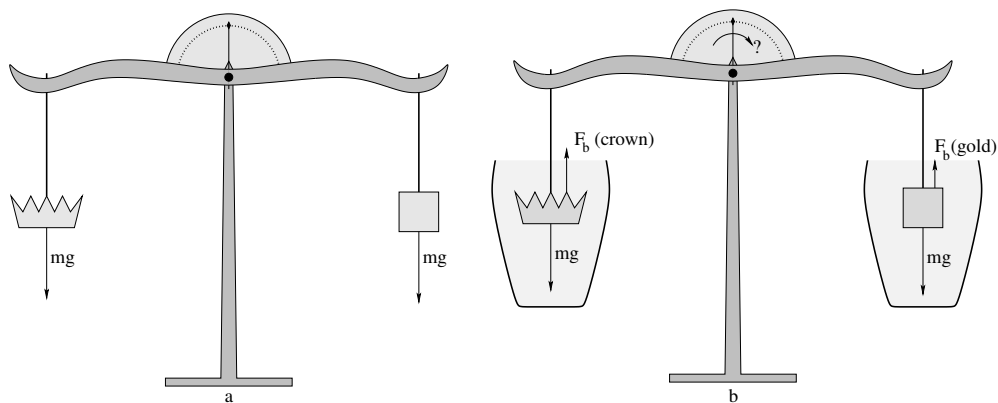


Figure 109: In a), the crown is balanced against an equal weight/mass of pure gold *in air*. In b) the crown *and* the gold are symmetrically submerged in containers of still water.

Archimedes very likely used his balance to first select and trim a piece of gold so it had *exactly the same weight as the crown* as illustrated in figure 109a. Then all he had to do was submerge the crown and the gold symmetrically in two urns filled with water, taking care that they are both fully underwater.

Pure gold is *more dense* than gold adulterated with silver (the most likely metal the goldsmith would have used; although a few others such as copper might have also been available and/or used they are also less dense than gold). This means that any given mass/weight (in air, with its negligible buoyant force) of adulterated gold would have a **greater volume** than an equal mass/weight (in air) of pure gold.

If the crown were made of pure gold, then, the buoyant forces acting on the gold and the crown would be equal. The weights of the gold and crown are equal. Therefore the submerged crown and submerged gold would be supported in static equilibrium by the *same force* on the ropes, and the balance would indicate “equal” (the indicator needle straight up). The goldsmith lives, the king is happy, Archimedes lives, everybody is happy.

If the crown is made of less-dense gold *alloy*, then its volume will be greater than that of pure gold. The buoyant force acting on it when submerged will therefore *also* be greater, so the tension in the string supporting it needed to keep it in static equilibrium will be smaller.

But this smaller tension then would fail to balance the *torque* exerted on the balance arms by the string attached to the gold, and the whole balance would rotate to the right, with the more dense gold sinking relative to the less dense crown. The balance needle would *not* read “equal”. In the story, it didn’t read equal. So sad – for the goldsmith.

Example 8.3.2: Testing the Crown II

Of course nowadays we don't do things with balance-type scales so often. More often than not we would use a *spring balance* to weigh something from a string. The good thing about a spring balance is that you can *directly read off the weight* instead of having to delicately balance some force or weight with masses in a counterbalance pan. Using such a balance (or any other accurate scale) we can measure and record the *density of pure gold* once and for all.

Let us imagine that we have done so, and discovered that:

$$\rho_{\text{Au}} = 19300 \text{ kilograms/meter}^3 \quad (8.61)$$

For grins, please note that $\rho_{\text{Ag}} = 10490 \text{ kilograms/meter}^3$. This is a bit over half the density of gold, so that adulterating the gold of the crown with 10% silver would have decreased its density by around 5%. If the mass of the crown was (say) a kilogram, the goldsmith would have stolen 100 grams – almost four ounces – of pure gold at the cost of 100 grams of silver. Even if he stole twice that, the 9% increase in volume would have been nearly impossible to directly observe in a worked piece. At that point the *color* of the gold would have been off, though. This could be remedied by adding *copper* ($\rho_{\text{Cu}} = 8940 \text{ kilograms/meter}^3$) along with the silver. Gold-Silver-Copper all three alloy together, with silver whitening and yellowing the natural color of pure gold and copper reddening it, but with the two *balanced* one can create an alloy that is perhaps 10% *each* copper and silver that has almost exactly the same color as pure gold. This would *harden* and *strengthen* the gold of the crown, but you'd have to damage the crown to discover this.

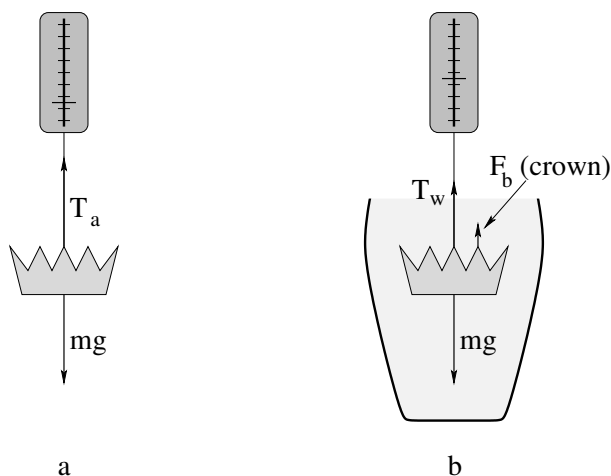


Figure 110: We now measure the effective weight of the *one* crown both in air (very close to its true weight) and in water, where the measured weight is *reduced* by the buoyant force.

Instead we hang the crown (of mass m) as before, but this time from a spring balance, both in air and in the water, recording both weights *as measured by the balance* (which measures, recall, the tension in the supporting string). This is illustrated in figure 110, where we note that the measured weights in a) and b) are T_a , the weight in air, and T_w , the measured weight while immersed in water.

Let's work this out. a) is simple. In static equilibrium:

$$\begin{aligned} T_a - mg &= 0 \\ T_a &= mg \\ T_a &= \rho_{\text{crown}} V g \end{aligned} \quad (8.62)$$

so the scale in a) just measures the almost-true weight of the crown (off by the buoyant force exerted by the *air* which, because the density of air is very small at $\rho_{\text{air}} \approx 1.2$ kilograms/meter³, which represents around a 0.1% error in the measured weight of objects roughly the density of water, and an even smaller error for denser stuff like gold).

In b):

$$\begin{aligned} T_w + F_b - mg &= 0 \\ T_w &= mg - F_b \\ &= \rho_{\text{crown}} V g - \rho_{\text{water}} V g \\ &= (\rho_{\text{crown}} - \rho_{\text{water}}) V g \end{aligned} \quad (8.63)$$

We know (we *measured*) the values of T_a and T_w , but we don't know V or ρ_{crown} . We have two equations and two unknowns, and we would like most of all to solve for ρ_{crown} . To do so, we *divide these two equations by one another* to eliminate the V :

$$\frac{T_w}{T_a} = \frac{\rho_{\text{crown}} - \rho_{\text{water}}}{\rho_{\text{crown}}} \quad (8.64)$$

Whoa! g went away too! This means that from here on we don't even care what g is – we could make these weight measurements on the moon or on mars and we'd still get the *relative* density of the crown (compared to the density of water) right!

A bit of algebra-fu:

$$\rho_{\text{water}} = \rho_{\text{crown}} \left(1 - \frac{T_w}{T_a}\right) = \rho_{\text{crown}} \frac{T_a - T_w}{T_a} \quad (8.65)$$

or finally:

$$\rho_{\text{crown}} = \rho_{\text{water}} \frac{T_a}{T_a - T_w} \quad (8.66)$$

We are now prepared to be precise. Suppose that the color of the crown is very good. We perform the measurements above (using a scale accurate to *better* than a hundredth of a Newton or we might end up condemning our goldsmith due to a *measurement error!*) and find that $T_a = 10.00$ Newtons, $T_w = 9.45$ Newtons. Then

$$\rho_{\text{crown}} = 1000 \frac{10.00}{10.00 - 9.45} = 18182 \text{ kilograms/meter}^3 \quad (8.67)$$

We subtract, $19300 - 18182 = 1118$; divide, $1118/19300 \times 100 = 6\%$. Our crown's material is around *six percent less dense than gold* which means that our clever goldsmith has adulterated the gold by removing some 12% of the gold (give or take a percent) and replaced it with some mixture of silver and copper. Baaaaad goldsmith, bad.

If the goldsmith were smart, of course, he could have beaten Archimedes (and us). What he needed to do is adulterate the gold with a mixture of metals that have *exactly the*

same density as gold! Not so easy to do, but tungsten's density, $\rho_W = 19300$ (to three digits) almost exactly matches that of gold. Alas, it has the highest melting point of all metals at 3684°K , is enormously hard, and might or might not alloy with gold or change the color of the gold if alloyed. It is also pretty expensive in its own right. Platinum, Plutonium, Iridium, and Osmium are all even denser than gold, but three of these are very expensive (even more expensive than gold!) and one is very explosive, a transuranic compound used to make nuclear bombs, enormously expensive *and* illegal to manufacture or own (and rather toxic as well). Not so easy, matching the density via adulteration and making a profit out of it...

Enough of all of this *fluid statics*. Time to return to some *dynamics*.

8.4: Fluid Flow

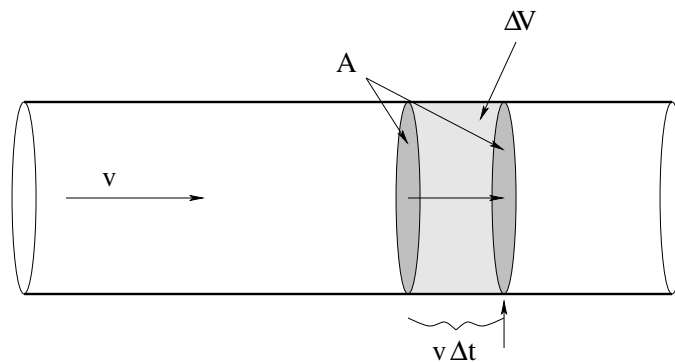


Figure 111: Fluid in uniform flow is transported down a pipe with a constant cross-section at a constant speed v . From this we can easily compute the *flow*, the volume per unit time that passes (through a surface that cuts the pipe at) a point on the pipe.

In figure [111](#) we see fluid flowing from left to right in a circular pipe. The pipe is assumed to be “frictionless” for the time being – to exert no drag force on the fluid flowing within – and hence all of the fluid is moving *uniformly* (at the same speed v with no relative *internal* motion) in a state of **dynamic equilibrium**.

We are interested in understanding the **flow** or **current** of water carried by the pipe, which we will define to be the **volume per unit time** that passes any given point in the pipe. Note well that we could instead talk about the *mass per unit time* that passes a point, but this is just the volume per unit time times the density and hence for fluids with a more or less *uniform* density the two are the same within a constant.

For this reason we will restrict our discussion in the following to incompressible fluids, with constant ρ . This means that the concepts we develop will work gangbusters well for understanding water flowing in pipes, beer flowing from kegs, blood flowing in veins, and even rivers flowing slowly in not-too-rocky river beds but not so well to describe the dynamical evolution of weather patterns or the movement of oceanic currents. The *ideas* will still be extensible, but future climatologists or oceanographers will have to work a bit harder to understand the correct theory when dealing the compressibility.